ZEBRA		Document Number	OB-WA-04-TX4
Title:	**TX4 Distribution Center Battery Management Process**	Revision	1
		Effective Date	
		Page:	Page **1** of **8**
Authors: Matthew Kelley			

BUSINESS UNIT: ☐ AIT ☐ EVM ☒ ALL

LOCATION(S): ☐ ALL (Global) ☒ NALA ☐ APAC ☐ EMEA

**Refer to Scope for possible limitations and/or elaborations relating to a selected region(s)*

DISTRIBUTION

Quality Assurance/Quality Control – QA
Material Control – MC
Global Planning & Purchasing – GP
Sourcing/Commodity Management/Procurement – SC
Manufacturing/Reconfiguration – MF
Distribution Center--DC

REVISION HISTORY

Rev	Description for change (*describe impact on process/product*)	Process impact/risk	Author	Date
1	Initial release. TX4 DC Battery management system		Matthew Kelley	7-12-2023

1. PURPOSE

1.1. To define a battery management process to restrict the shipment from TX4/5 to customers and partners, of rechargeable battery packs that do not meet Zebra's battery quality standards and requirements. This procedure will outline the process to periodically purge battery stock from inventory and perform testing to ensure product in stock is meeting quality standards prior to shipping.

2. SCOPE

2.1. This procedure is applicable to the TX4/5 distribution center and defines the minimum requirements for managing battery inventory.
2.2. Applies to both EVM and ATS batteries.
2.3. Applies to batteries received from battery pack suppliers.
2.4. Applies to batteries returned through the Stock Rotation process (re-stockable items).
2.5. Limited to lithium-ion batteries, this excludes Lithium metal, nickel metal, lead acid batteries and coin cells.
2.6. Not applicable to batteries packaged with product.

3. REFERENCES

3.1. Battery Test Procedure OB-TE-01 (ISO Library) OB-TE-01
3.2. Battery Management Procedure OB-WA-04 (ISO Library) OB-WA-04
3.3. Inventory Scrap Policy FP3.4 (zConnect, Finance page) FP3.4: Inventory Scrap
3.4. Delegation of Authority Policy GP2 (zConnect, Finance page) GP2 - Delegation of Authority
3.5. Control of Quality and Environmental Management System Records OZ-CQ-09 (ISO Library) OZ-CQ-09
3.6. Site Specific Procedures for Nonconforming Material, Inventory management and Purge, Data Collection, MRB, Scrap, and Damaged Packaging

4. DEFINITIONS / ABBREVIATIONS / ACRONYMS

4.1. DC: Distribution Center
4.2. ATS: Asset Identification and Tracking including printers, printing supplies and location solutions.
4.3. EVM: Enterprise Visibility and Mobility including mobile computers, and scanning solutions
4.4. Stock Rotation: Product returned from distributors and deemed acceptable for restock
4.5. Shelf Life: Defined length of time that batteries are considered acceptable to ship
4.6. Date Code: Defines the Manufacture date of the battery pack on the overpack (a package of individual packages of batteries) not the individual battery pack. Date codes are in various formats (see Appendix A)
4.7. MRB: Material Review Board (Cross functional team that determines disposition of suspect material)
4.8. RMA: Return Material Authorization
4.9. Battery Lot: A group of batteries with the same inventory part number, and battery pack label part number manufactured on the same day by the same manufacturer.
4.10. Inter DC shipments: Shipments between DC's for example a shipment from NL1 to TX4.
4.11. WIP Job: Work In Progress Job
4.12. Sub-inventory REW2: TX4/5's Rework locator, non-standard WIP Job are always released from REW2.
4.13. Sub-inventory RMF1: Re-Configuration sub-inventory, WIP Jobs need to be dropped to this sub-inventory to start the actual job.
4.14. Sub-inventory FIG1: Finished goods sub-inventory.
4.15. OAC: Oracle Analytics Cloud is a platform to access and process data.
4.16. OPS: Operations
4.17. IC: Inventory Control

5. RESPONSIBILITIES

5.1. **Design and Development Engineering:** Provide and update Cadex Battery Analyzer Test files and Zebra ECRT.

5.1.1. **AIT:** Design & Development Engineering (Frank Calabresi)

5.1.2. **EVM:** Electrical Engineering (Someone under David Stern)

5.1.3. **Testing Equipment:** Test Engineer (Jesse Abruzzo)

5.2. **Test Engineering:** Calibrate and maintain test equipment including Cadex battery analyzers, adapters, and automated data-gathering and voltage-measurement units.

5.2.1. System / Test Engineer for calibration (Sanjay Patel)

5.3. **Test Engineering:** Train test equipment users.

5.4. **Operations:** Receive battery lots from battery pack manufacturers.

5.5. **Operations:** Receive battery lots from channel partners and inter-DCs battery transfers.

5.6. **Quality:** Determine battery lot date of manufacture through OAC.

5.7. **Inventory Planning:** Execute scrap procedures for batteries manufactured more than three years ago.

5.8. **Quality:** Add Sub-inventory move request to SharePoint list to batteries moving to REW2

5.9. **Inventory Control:** Move batteries to REW2 and place Locator Status: Zebra REW2 in all FIG1/FWP1 locators.

5.10. **Inventory Planning:** Place work order for batteries in REW2 needing testing.

5.11. **Operations:** Pick material outline in work order in REW2 to RMF1.

5.12. **Operations:** Execute battery lot acceptance test according to OB-TE-01.

5.13. **Operations:** Return to stock in original packaging those lots that are deemed acceptable based on testing criteria; exclusive of those batteries which were capacity-tested – these should be scrapped

5.14. **Inventory Control:** Remove Locator Status: Zebra REW2 on the original location the batteries returning to stock after testing were in.

5.15. **Inventory Control:** communicate scrapped batteries to inventory planning.

5.16. **Inventory Control:** Scrap batteries out of the system according to FP3.4.

5.17. **Quality:** For lots deemed unacceptable, call a Material Review Board meeting to disposition material.

5.18. **Quality:** Initiate the process to scrap batteries and lots deemed failed by the Material Review Board according to 6.5.1 OB-WA-04.

5.19. **Quality:** communicate scrapped batteries to inventory planning.

5.20. **Operations and Test Engineering:** Archive test result records (battery aging report Excel spreadsheet.

5.21. **Operations:** Ensure that batteries that passed acceptance testing and those received from battery manufacturers are stored in compliance with OB-WA-04 section 6.3.

5.22. **Operations:** A FIFO logic is used for shipment of batteries from the Distribution Center based preferably on date of manufacture but acceptably on date of arrival.

5.23. **Quality:** Audit process at each DC Quality including verifying age report is up-to-day.

5.24. **Quality and Operations:** Shall employ a Battery Storage Control Plan to ensure the requirements of Section 6.3 are met.

5.25. **Quality:** Archive results 7.1 OB-WA-04

6. PROCESS

6.1. Incoming (Receiving)

6.1.1. Receive in batteries from supplier according to work instructions.

6.1.2. Receive in internal shipments according to work instructions.

6.1.2.1. E-mail quality to get rid of inspection flag.

6.1.2.2. Follow quality's advice (either move to FIG1/FWP1 or REW2).

6.2. DC Incoming (Return Material Authorization – Stock Rotations)

6.2.1. Battery Lots returned from Zebra partners must be checked for manufacturing date before being placed in inventory. This date must be used to determine if the Lot is subject to testing.

6.3. Put Away

6.3.1. Supplier batteries move to FIG1/FWP1

6.3.2. Internal & Stock Rotation batteries move to FIG1/FWP1 if younger than 1 year or move to REW2 if older than 1 year.

6.4. Monthly Report

6.4.1. Quality will run a monthly report from OAC.

6.4.2. Quality will determine impacted (according to OB-WA-04) battery lots.

6.4.3. Inventory Control will move impacted battery lots that Quality determined to REW2.

6.4.4. Quality will communicate battery testing results & age with planning, OPS & IC on an as needed basis through MRB meeting, NCR meeting, and/or through email.

6.5. Raise WIP Job

6.5.1. Planner will raise WIP Job with pick release for battery testing.

6.6. Pick Material

6.6.1. Picker will pick material and physical drop in Re-Configuration Staging area outside the fence.

6.7. Drop Material & Run Test

6.7.1. Re-Configuration operator will drop material in system to RMF1.

6.7.2. Re-Configuration operator will perform test according to OB-TE-01.

6.7.3. Test results will be automatically recorded in Splunk.

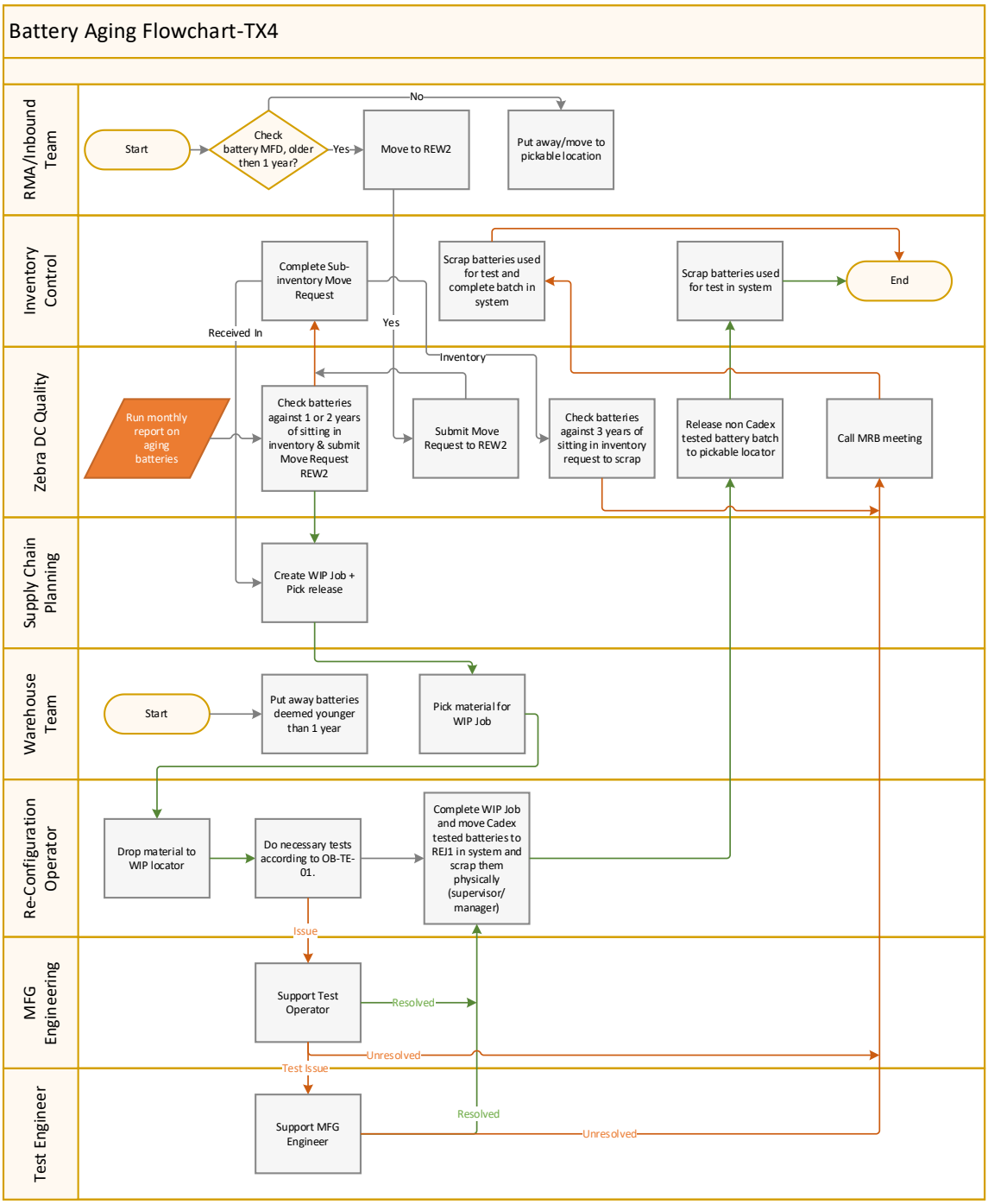
6.8. Scrap Tested Batteries

6.8.1. Re-Configuration operator with Supervisor/Manager will scrap batteries according to waste management.

6.9. **Release/Scrap Stock**

6.9.1. If tested passed quality will release stock and tested lot will be moved to stock, if failed quality will call a MRB meeting to discuss disposition according to OB-WA-04 6.4.4.

APPENDIX A


Battery Aging Flowchart-TX4
RMA/Inbound Team
Start
Check battery MFD, older then 1 year?
No
Yes
Move to REW2
Put away/move to pickable location
Inventory Control
Complete Sub-inventory Move Request
Scrap batteries used for test and complete batch in system
Scrap batteries used for test in system
End
Received In
Yes
Inventory
Zebra DC Quality
Run monthly report on aging batteries
Check batteries against 1 or 2 years of sitting in inventory & submit Move Request REW2
Submit Move Request to REW2
Check batteries against 3 years of sitting in inventory request to scrap
Release non Cadex tested battery batch to pickable locator
Call MRB meeting
Supply Chain Planning
Create WIP Job + Pick release
Warehouse Team
Start
Put away batteries deemed younger than 1 year
Pick material for WIP Job
Re-Configuration Operator
Drop material to WIP locator
Do necessary tests according to OB-TE-01.
Complete WIP Job and move Cadex tested batteries to REJ1 in system and scrap them physically (supervisor/ manager)
Issue
MFG Engineering
Support Test Operator
Resolved
Unresolved
Test Issue
Test Engineer
Support MFG Engineer
Resolved
Unresolved